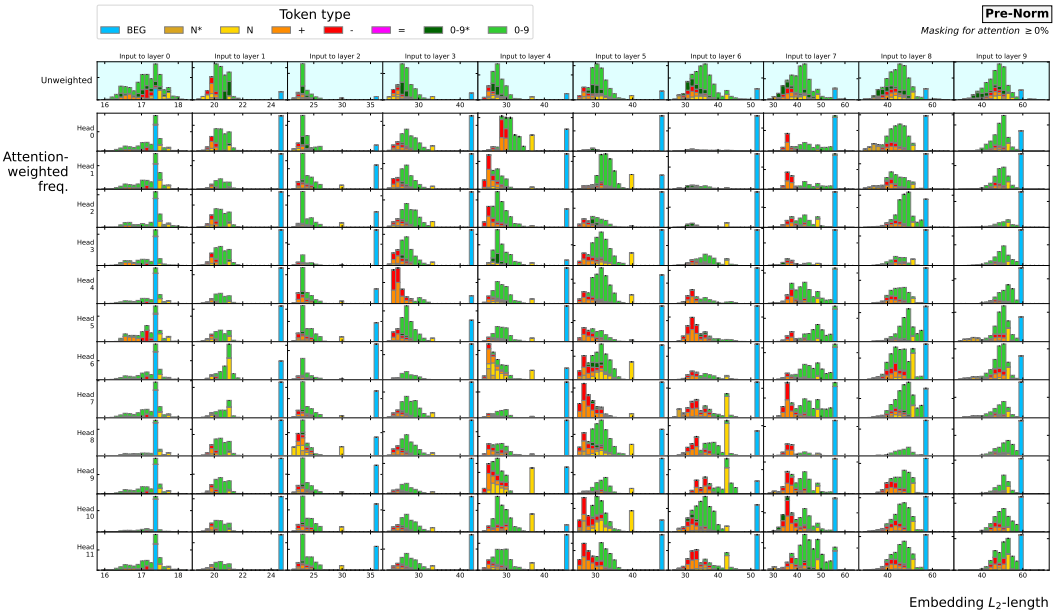

Figure 11: Embedding distributions at different model depths using the `Baseline Pre-Norm` model. The categories `0-9` and `N` are separated into whether they occur in the question (light colour) or answer (dark colour, distinguished by $*$ label). These distributions are used to compute the LHS of Figure 1 after removing the `BEG` tokens.

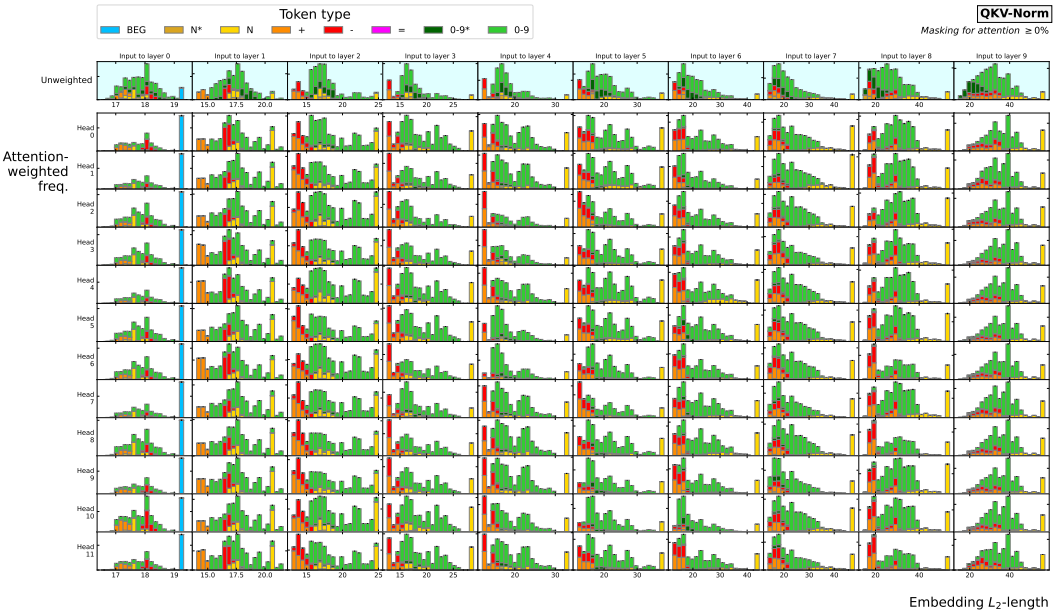

Figure 12: Embedding distributions at different model depths using the `Baseline QKV-Norm` model. The categories `0-9` and `N` are separated into whether they occur in the question (light colour) or answer (dark colour, distinguished by $*$ label). Note that overflow panels are excluded from this plot for legibility. These distributions are used to compute the RHS of Figure 1 after removing the `BEG` tokens.

B.2 Circuit collapse

Figure 3 shows the probability of circuit collapse. This is the probability that an attention distribution with no noise selects embedding i with high probability $a_i \geq 95\%$, and when noise is added, it transitions such that some $k \neq i$ becomes the maximum attended embedding. This definition is chosen because it matches our theoretical results in section 6. However, it does not require that the distribution remains sparse after the noise addition. Figure 13 compares this baseline result (top) with an alternative definition (bottom), in which the second distribution must also be sparse, meaning $a_k \geq 95\%$ for some $k \neq i$. We see that 1% of sparse attention distributions collapse at a noise level of 11% when using the original definition, delayed until 17% when using the sparse definition. Therefore we observe that the sparse-to-sparse case does occur, but requires a higher noise level.

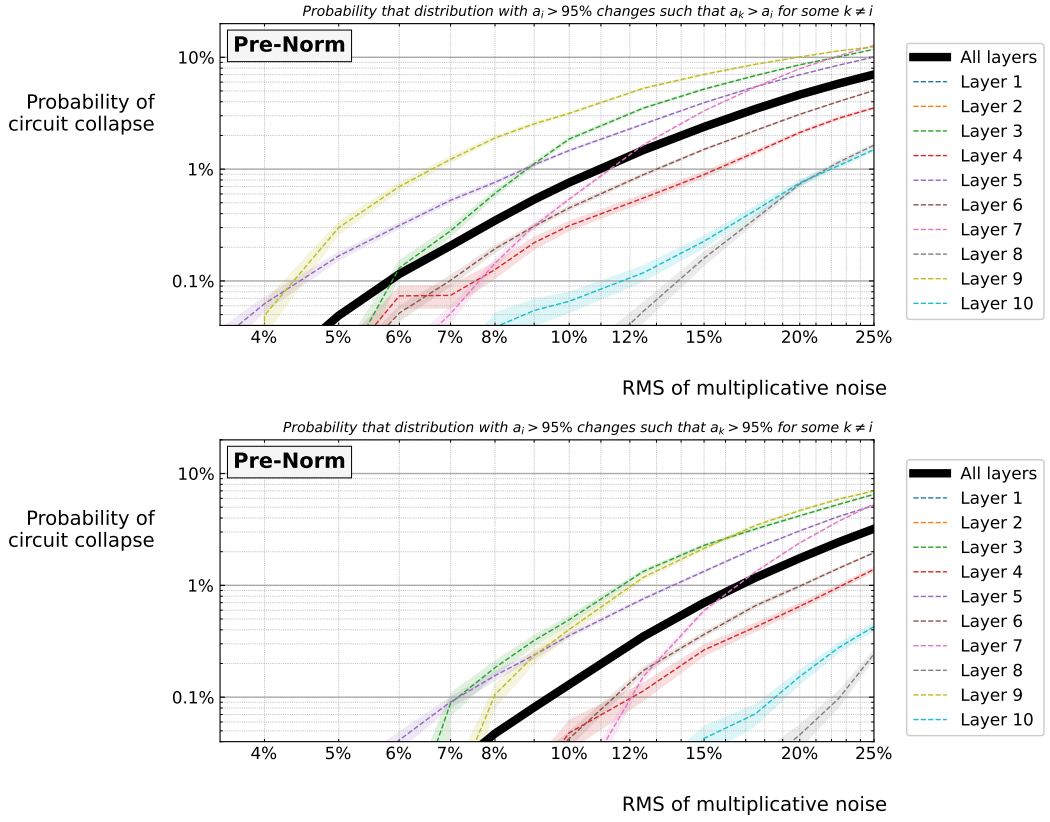

Figure 13: Probability of circuit collapse vs increasing noise. **Top:** using the baseline definition. This is a reproduction of Figure 3. **Bottom:** requiring the attention distribution to remain sparse after switching to a different token.

C Main experiments: results with different models

In this appendix we reproduce the main experimental results using our model variations.

C.1 Embedding lengths

Figure 1 shows the empirical results demonstrating the attention-weighted spread of embeddings. Figures 14-16 show the results we obtain when we perform the same analysis using the `Alternate` and `Large` model variations. In all cases, we observe 90% of embeddings within a spread of roughly $\pm 30\%$ when using `Pre-Norm`. In all cases, the spread of embeddings for `QKV-Norm` is larger, although we note that the effect is smaller when using the `Alternate` variation.

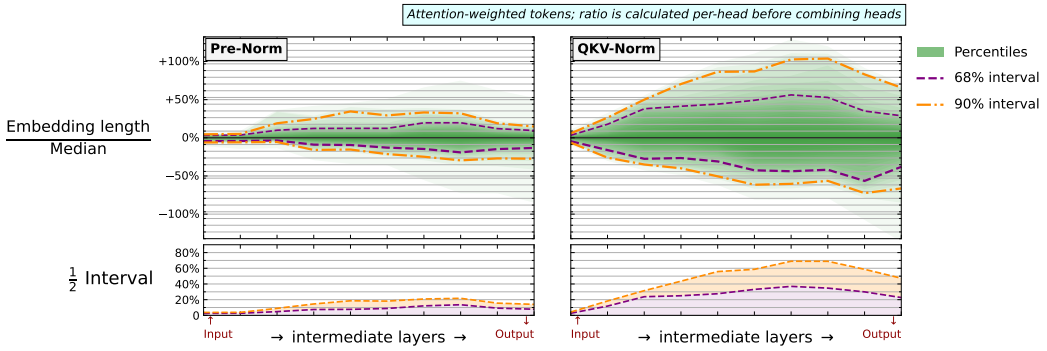

Figure 14: Attention-weighted spread of embeddings at increasing model depth using the `Baseline` model and task configuration. This is a replication of Figure 1.

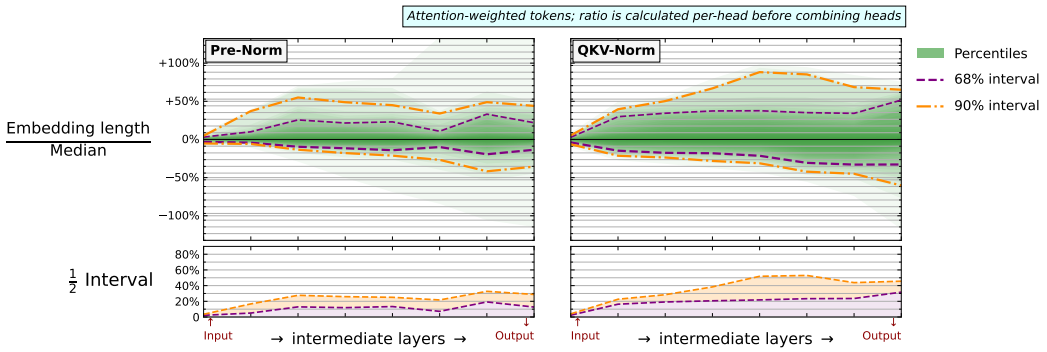

Figure 15: Attention-weighted spread of embeddings at increasing model depth using the `Alternate` model and task configuration.

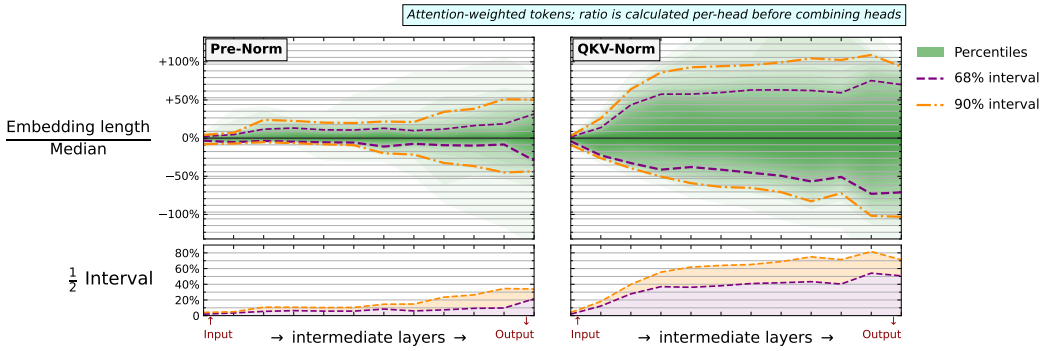

Figure 16: Attention-weighted spread of embeddings at increasing model depth using the `Large` model and task configuration.